Test Summary and Defect Report

CSE 6324 Advanced Topics in Software Engineering · Team 3 · MavAssist

Generated 2026-11-10 by scripts/build_test_report.py from the raw results in test-evidence-final-2026-11-10/ and test-evidence-2026-11-10/. Every number below is read from a raw result file; nothing was entered by hand.

1. Scope and environment

Scope: the MavAssist web application (student directory, search, assistant, saved and recent resources; administrator resources, categories, notices and usage insights), its API, and the Docker stack with PostgreSQL. Out of scope this term: cloud staging and BrowserStack (see docs/SCOPE_CHANGES.md).

Item	Environment
Machine	Windows 11 laptop; Docker Desktop for the container stack
Backend tests	pytest with an isolated SQLite database and AI keys blank (as in CI); coverage 94%
Browser tests	Playwright on Chromium, Firefox and WebKit against the API in offline mode and the web dev server
Load tests	k6, 50 virtual users for 60 s: local API on SQLite, and the Docker stack on PostgreSQL (AI keys blank)
Security scan	OWASP ZAP baseline (passive) against the Docker web app and API
AI routing	Offline: keyword and semantic retrieval with cited sources. Hosted: diagnostic sample only (provider limits)
Runs	Nov 9 baseline and Nov 10 retest by Harmanpreet Kaur; Nov 10 re-run and final run by Ashwinder Singh

2. Results by test type

Test type	Total	Pass	Fail	Blocked	Descoped	Not run	Pass rate
Security / role	3	3	0	0	0	0	100%
Backend API integration	41	41	0	0	0	0	100%
Backend guardrail	71	71	0	0	0	0	100%
Component	1	1	0	0	0	0	100%
Unit	2	2	0	0	0	0	100%
Build / type check	1	1	0	0	0	0	100%
End-to-end	9	9	0	0	0	0	100%
Accessibility / responsive E2E	3	3	0	0	0	0	100%
Mobile E2E / 360 px	3	3	0	0	0	0	100%
Local smoke	2	2	0	0	0	0	100%
Docker smoke	1	1	0	0	0	0	100%
Docker live	1	1	0	0	0	0	100%
Performance	4	1	2	0	0	1	33%
Performance diagnostic	1	1	0	0	0	0	100%
Security scan	2	2	0	0	0	0	100%
AI routing (offline)	2	2	0	0	0	0	100%
AI routing (hosted)	2	1	0	1	0	0	100%
Usability study	2	0	0	0	0	2	n/a
AI dataset review	1	0	0	0	0	1	n/a
Deployment	1	0	0	0	0	1	n/a
Cross-browser (cloud)	1	0	0	0	1	0	n/a
Total	154	145	2	1	1	5	99%

Pass rate is passed / (passed + failed); blocked, descoped and not-run cases are counted separately.

3. Requirement coverage

Req.	Description	Tests	Pass	Status	Reason
FR-01	Student registration and login; administrator role gate	8	8	Met	
FR-02	Service directory and detail (location, hours, contact, source date)	6	6	Met	
FR-03	Keyword search and category filtering	4	4	Met	
FR-04	Assistant recommends the right office with reason, cited steps and clarification	112	110	Partial	Offline routing passes; full hosted evaluation blocked by provider limits (D-12); label review not recorded.
FR-05	Administrator create, update, archive and restore services	6	6	Met	

FR-06	Categories and date-bounded notices	5	5	Met	
FR-07	Bookmarks	4	4	Met	
FR-08	Recently viewed (ten most recent)	4	4	Met	
FR-09	Usage analytics by date range	5	5	Met	
FR-10	Helpful / not-helpful answer ratings	4	4	Met	
NFR-01	Performance: 50 users, non-AI p95 < 1 s, AI p95 < 5 s, errors < 1%	7	4	Partial	Local SQLite load test passes; Docker/PostgreSQL latency and hosted answer latency miss the targets (D-15); staging not run.
NFR-02	Security: password hashing, token expiry, role checks, headers, HTTPS	11	10	Partial	Role tests and ZAP baseline pass; HTTPS on cloud staging not verified (local Docker demo).
NFR-03	Privacy: no raw questions or identifiers stored; redaction; retention	4	4	Met	Privacy tests pass. Hosted answers send a redacted question plus public office context (clarification in REQUIREMENTS.md).
NFR-04	Usability and accessibility: WCAG AA, responsive 360 px, SUS >= 70	14	11	Partial	Accessibility and 360 px checks pass on three engines; SUS study not run yet.
NFR-05	Reliability: fallback when the hosted model fails; coverage >= 70%	9	8	Partial	Not passing: STAGE-001 (Not run)

4. Performance

Run (50 users × 60 s)	Requests	Errors	Non-AI p95	Chat p95	Result
Local API, SQLite, offline mode	3832	0%	192 ms	639 ms	Pass
Docker stack, PostgreSQL, AI keys blank	648	0%	1,893 ms	12,026 ms	Fail

Targets: errors < 1%, non-AI p95 < 1,000 ms, chat p95 < 5,000 ms. The Docker stack had no errors but missed both latency targets; it runs semantic retrieval on the CPU for every query, the likely cause (not yet confirmed). Hosted answers (diagnostic, 63 answers) took p50 6.3 s and p95 15.8 s against a 5 s target (D-15). A per-question time limit that falls back to cited summaries is implemented; its before/after measurement is pending.

5. AI routing accuracy

Set (offline, measured)	Questions	Top-1	Top-3	Unknown offices	Result
100-question set	100	97%	100%	0	Pass
50-question holdout	50	90%	100%	0	Pass

Hosted model, noted separately: a diagnostic sample of 63 hosted answers on the 100-question set, all 63 correct at top-1, before the paid account ran out of credit. This is a diagnostic, not acceptance: the full hosted evaluation (AI-003) is blocked by provider limits (D-12).

6. Security scan

Target	High	Medium	Low	Info
Web app, before fix	0	1	5	2
Web app, after fix	0	0	5	2
API	0	0	0	1

The medium finding (Content-Security-Policy without form-action, which has no fallback) was fixed in nginx.conf and vercel.json and re-scanned. Remaining low findings are documented in docs/SECURITY_SCAN.md.

7. Defects

ID	Sev.	Found	Summary	Status	Commit	Verified by
D-01	High	2026-10-23	"someone mollested me" answered "That's not something MavAssist can help with"	Fixed	a8c1032 (PR #7)	BE-102, BE-104
D-02	High	2026-10-23	Complaint about racist treatment by a staff member at "accounts" was routed to Student Accounts	Fixed	a8c1032 (PR #7)	BE-065
D-03	High	2026-10-23	"i wanna hit someone" answered "Not found"	Fixed	a8c1032 (PR #7)	BE-095

D-04	Medium	2026-10-23	"i got scammed" answered "Not found"	Fixed	a8c1032 (PR #7)	BE-100
D-05	Low	2026-10-23	Insults ("fuck you") answered "Needs a detail"	Fixed	fc26feb (PR #4)	BE-038, BE-067
D-06	Low	2026-10-23	Scroll arrows visible inside the one-line question box	Fixed	fc26feb (PR #4)	visual check: test-evidence-final-2026-11-10/screenshots/assistant-empty-desktop.png
D-07	Medium	2026-10-23	API container exits with "password authentication failed" after unzipping a new copy	Fixed	a8c1032 (PR #7)	DOCKER-001
D-08	Medium	2026-10-23	Browser check job failing on the assistant pull request	Fixed	a8c1032 (PR #7)	E2E-002, E2E-007, E2E-012
D-09	Medium	2026-11-09	Serious contrast failures: drawer text 4.15:1, verified badge 3.71:1, example chip 2.50:1	Fixed	a8c1032 (PR #7), 73e1bc1 (PR #6)	E2E-001, E2E-002, E2E-006, E2E-007, E2E-011, E2E-012
D-10	High	2026-11-09	52 of 3,100 requests failed (1.68%, target < 1%); slowest 30.8 s	Fixed	a8c1032, 0f0f407 (PR #7)	PERF-001
D-11	Medium	2026-11-09	Docker smoke and live tests blocked on the macOS test machine (Docker daemon not running)	Fixed	No code change; retest in 73e1bc1 (PR #6)	DOCKER-001, DOCKER-002
D-14	Low	2026-11-09	Adding a second browser spec made the unit test command load browser specs	Fixed	73e1bc1 (PR #6), 0f0f407 (PR #7)	FE-UNIT-001, FE-UNIT-002, FE-UNIT-003
D-12	High	2026-11-09	all_hosted_rag is false in every stored evaluation file; hosted accuracy not yet recorded	Open	91414be, 72933dd	AI-003, AI-004
D-13	High	2026-11-09	docs/usability-results.csv has no participant rows	Open	n/a	USAB-001
D-15	High	2026-11-10	Hosted answers take p50 4.9–6.3 s, p95 10.5–17.2 s (gpt-4o-mini, three runs); NFR-01 asks for chat p95 < 5 s	Open	03b3835	AI-005, BE-113

8. Open items and owners

Item	Owner	Next step
D-13 / USAB-001-002: SUS study with 10+ students	Krisha Patel	Run sessions per docs/SUS_STUDY.md; score with scripts/sus_score.py
LABEL-001: independent review of evaluation labels	Harmanpreet Kaur / Suhas S	Record decisions in docs/LABEL_REVIEW.md
STAGE-001-002: cloud staging and HTTPS	Ashwinder Singh (decision)	Deploy or confirm the local Docker demo (docs/SCOPE_CHANGES.md)
D-12 / AI-003: full hosted evaluation	Ashwinder Singh	Run in free-tier batches (evaluate_ai.py --sample) once a free model works
D-15 / AI-005 / PERF-003: latency	Ashwinder Singh	Measure the time limit before/after; investigate Docker retrieval cost

9. Selected screenshots

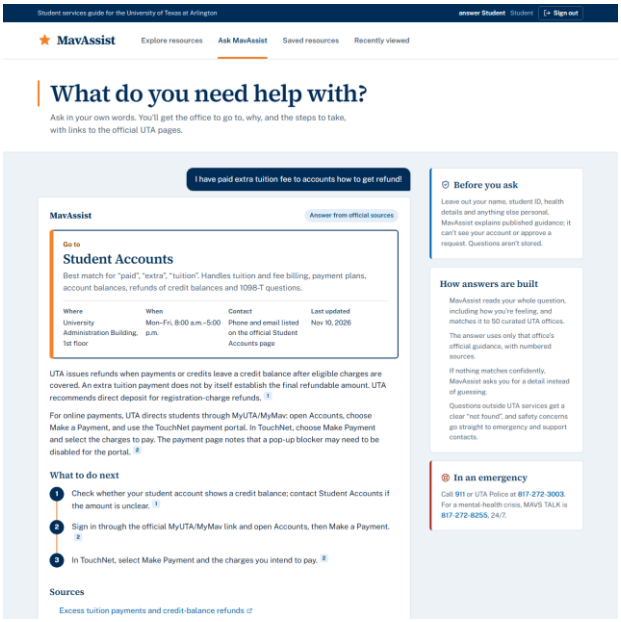


Figure 1. Assistant answer with cited sources

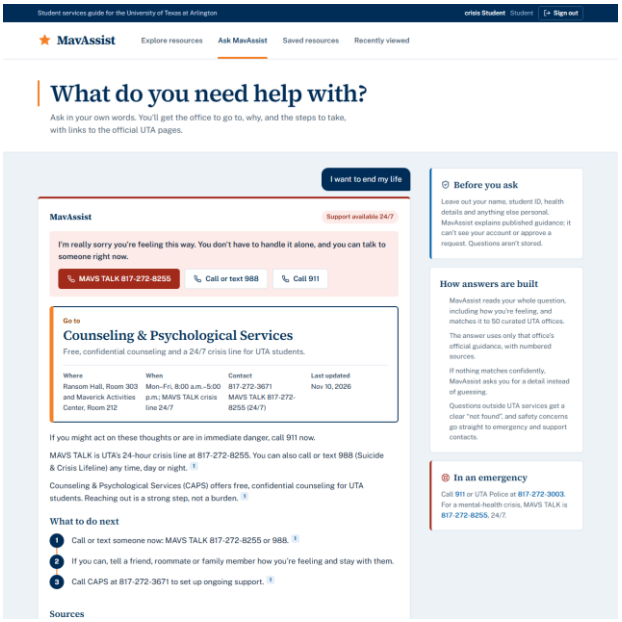


Figure 2. Crisis support response

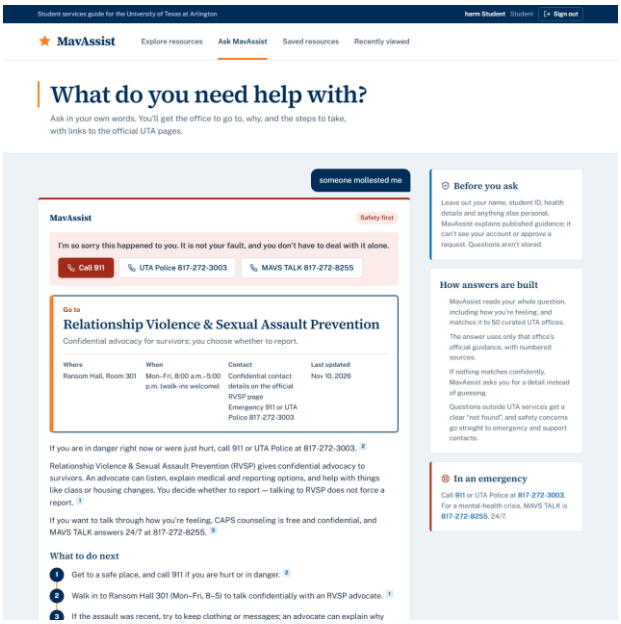


Figure 3. Harm report: victim support

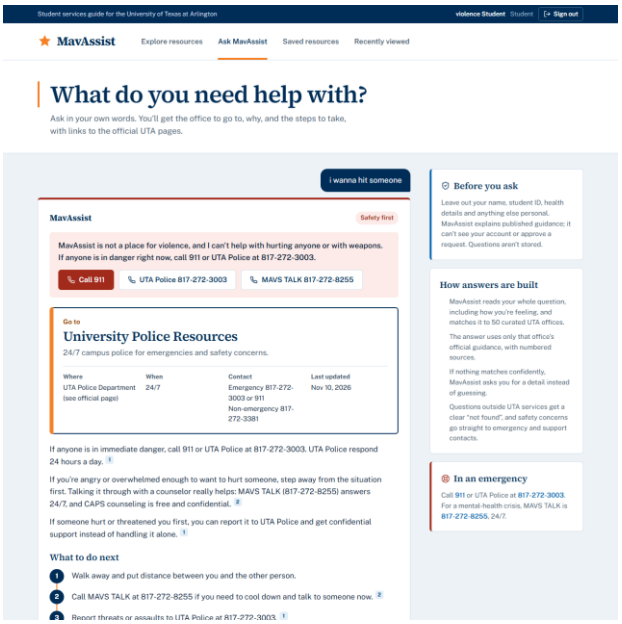


Figure 4. Violence: refusal and safety contacts

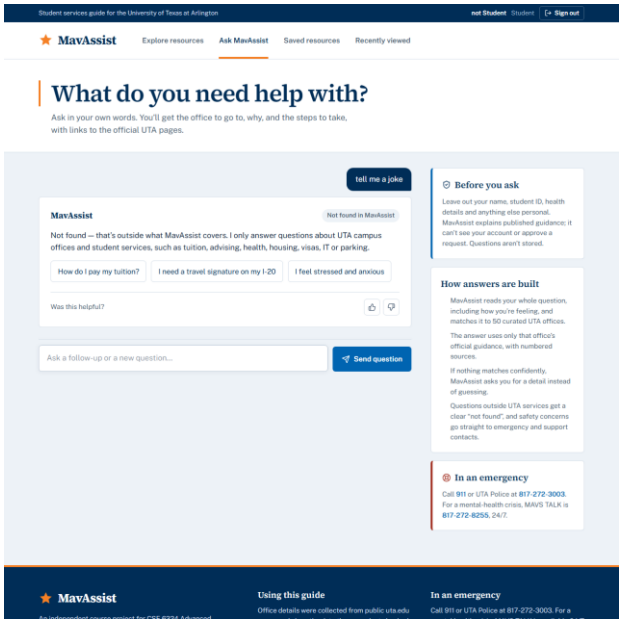


Figure 5. Out of scope: not found

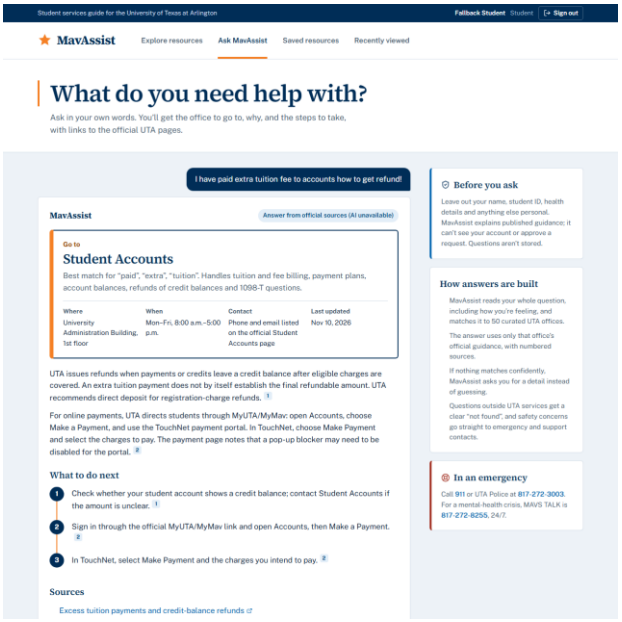


Figure 6. Hosted model unavailable: cited-source fallback

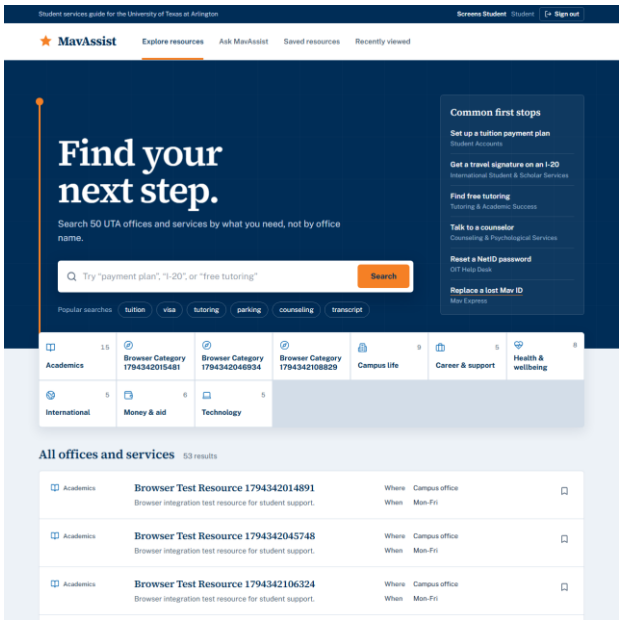


Figure 7. Student directory

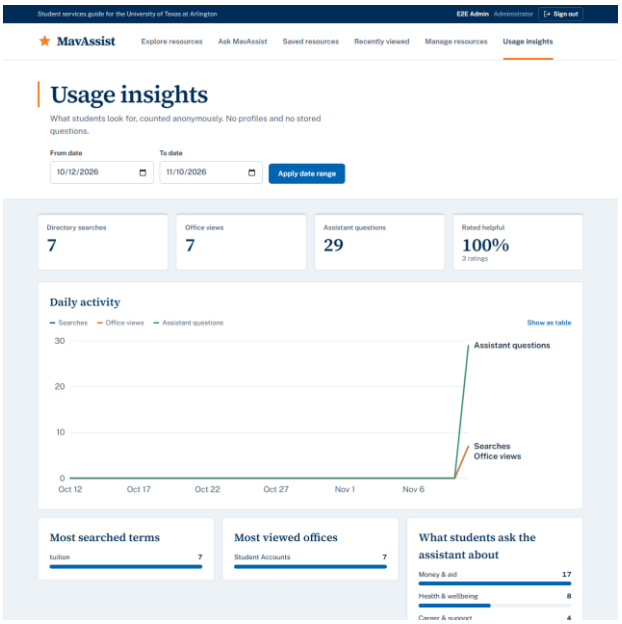


Figure 8. Administrator usage insights

All screenshots (desktop, 360 px, administrator pages) are in test-evidence-final-2026-11-10/screenshots/.